# Choose axes that help explain the ramp

GUIDED NOTES / use with the projected diagram Name _____

## A SAY THE WORD

### Weight components / ES: componentes del peso / PT: componentes do peso

mg sin θ along; mg cos θ into the ramp.

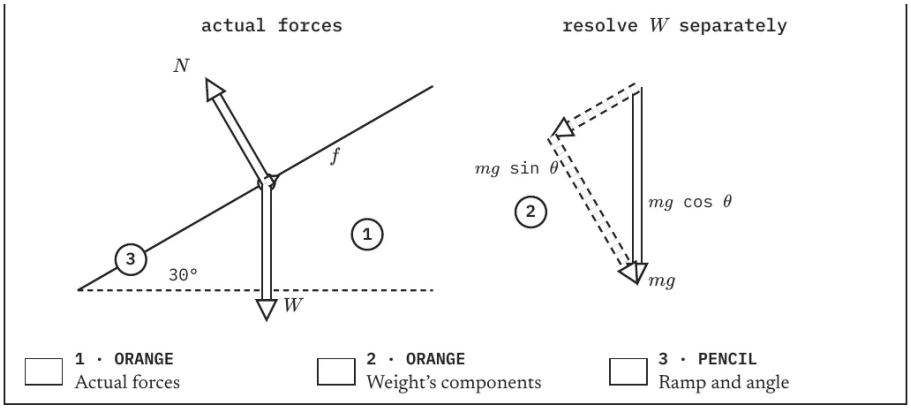
Also on your note page: Parallel axis, Perpendicular axis, Constraint.

## B READ THE RELATIONSHIP

### Downhill: ma = mg sin θ − f. Perpendicular: N = mg cos θ here.

As you read, find: parallel · perpendicular · downhill.

## C TRACE THE ORIGINAL FIGURE




Use the numbered key. Trace one feature. Tell your partner what it means.

## D FOLLOW THE WORKED EXAMPLE

### A 2 kg block slides down a 30° ramp with 2 N friction uphill. Find its acceleration and normal force. Use g = 9.8 m/s² .

**Given** m = 2 kg ; θ = 30° f = 2 N uphill a ⊥ = 0

**Rule** mg sin θ − f = ma ∥ N = mg cos θ

**Work** a ∥ = [19.6(0.5) − 2]/2 N = 19.6 cos 30°

**Answer** 3.9 m/s² Downhill; N ≈ 17.0 N .

## E SAVE FOR THE FINAL MEETING

A suitable set of axes can simplify the forces changing XJ's path. Components are calculation tools; they must not be counted again as extra forces.

# Read the physics: Weight components

SUPPORTED READING / read one part, then point to the evidence Name

## 01 PICTURE IT

On a ramp, gravity still points straight down. The normal force points straight out of the surface. Choose one axis along the ramp and one perpendicular to it.

## 02 HOW IT WORKS

Split weight into two parts. The parallel part pulls downhill; the perpendicular part presses into the ramp. Calculate with the parts or with the original weight, never with both together.

## 03 WHAT TO CHECK

Subtract uphill friction from the downhill force component. If the block stays on the ramp, its acceleration perpendicular to the ramp is zero.

## 04 TRY THE EXAMPLE FROM YOUR NOTES

A 2 kg block slides down a 30° ramp with 2 N friction uphill. Find its acceleration and normal force. Use $g = 9.8\ m/s^2$ . 3.9 $m/s^2$ Downhill; $N \approx 17.0$ N .

**Model check: Downhill: $ma = mg \sin\theta - f$. Perpendicular: $N = mg \cos\theta$ here.**

## 05 CONNECT TO XJ

A suitable set of axes can simplify the forces changing XJ's path. Components are calculation tools; they must not be counted again as extra forces.

### SAY IT BEFORE YOU WRITE

Why would adding both weight and its components be a mistake? Along the ramp I use ___. Across the ramp I use ___.

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Parallel axis

ES: eje paralelo / PT: eixo paralelo

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain parallel axis to a friend?<br>In my words, this means ___. | If downhill is positive, what sign does the along-ramp part of gravity have?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| Sketch axes for a ramp and mark the parallel axis.<br>My example shows ___. | A partner uses the horizontal floor as the parallel axis. Which surface should they follow?<br>I would change ___ because ___. |

## Perpendicular axis

ES: eje perpendicular / PT: eixo perpendicular

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain perpendicular axis to a friend?<br>In my words, this means ___. | If the positive axis points away from the ramp, what sign does the normal force have?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| Draw the perpendicular axis for a block on a tilted ramp.<br>My example shows ___. | A student draws that axis straight up on a tilted ramp. Why would you rotate it?<br>I would change ___ because ___. |

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Weight components

ES: componentes del peso / PT: componentes do peso

| Meaning in my words | How can you tell? |
|---|---|
| What are the two components of weight on a ramp?<br>In my words, this means ___. | Which part of weight pulls downhill, and which presses into the ramp?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| Draw weight and its two parts. Which arrows should enter the two ramp-axis equations?<br>My example shows ___. | A student adds weight and both of its components as three forces. What got counted twice?<br>I would change ___ because ___. |

## Constraint

ES: restricción / PT: restrição

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain constraint to a friend?<br>In my words, this means ___. | A block stays on a straight ramp. What is its acceleration perpendicular to the surface?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| How would you use that fact to find the normal force when gravity and the surface are the only forces?<br>My example shows ___. | Does zero perpendicular acceleration mean zero downhill acceleration?<br>I would change ___ because ___. |

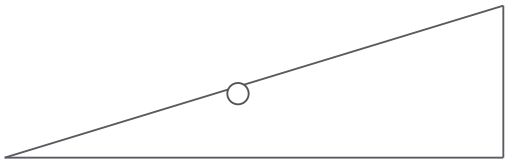
# Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

## Draw the real weight once, then draw its two components in a separate box. Circle the component along the ramp.

Labels you can use: parallel · perpendicular · downhill.

Real forces



Components

**Tell a partner**

Along the ramp I use ___. Across the ramp I use ___.

# On assignment: Weight components

SECTION E / BACK / FINAL MEETING ONLY Name _____

## A navigation graphic lists a gravitational force AND its two components as three separate pulls on XJ.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Navigation engineer**

Build an overlay using the component view. Indicate the full arrow it replaces before releasing the values to the calculator.

☐ **Map graphics editor**

Sketch two camera views: the real pull, then its components. Give viewers a caption that prevents counting the same pull twice.

**Word bank: parallel · perpendicular · downhill.**

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

**“We first said ___. Now we ___ because ___.”**

Editor’s check: State which component can change speed at that instant.